FINAL REPORT

Global Air Quality Analytics Pipeline

An end-to-end data analytics pipeline

By	Muhammad Aliyan & Amro Ahmed

Date	April 5, 2026
Tools Used	Microsoft Excel · Python (pandas + scikit-learn) · SQLite · Tableau · GCP
Datasets	Global Air Quality Dataset: https://www.kaggle.com/datasets/waqi786/global-air-quality-dataset Human Development Dataset: https://hdr.undp.org/data-center/human-development-index#/indicies/HDI GDP Dataset: https://data.worldbank.org/indicator/NY.GDP.PCAP.CD
GitHub Repo Link	https://github.com/aliyanm564/air-quality-analytics

ETL / Cleaning → SQL Storage → EDA → Prediction → Dashboard

Table of Contents

1. Introduction and Analytics Question
2. Dataset Description
3. ETL and Data Cleaning (Excel)
4. Data Storage (Python + SQLite)
5. Exploratory Data Analysis (Python)
6. Prediction Model (Python + scikit-learn)
7. Visualization and Dashboard (Tableau)
8. Results and Discussion
9. Limitations and Next Steps
10. References

1. Introduction and Analytics Question

This project implements an end-to-end data analytics pipeline using a public global air quality dataset. The pipeline demonstrates the full workflow from data acquisition through ETL, storage, exploratory analysis, predictive modeling, and interactive reporting. Three primary tools are used across the pipeline: Microsoft Excel for data cleaning and documentation, Python (with pandas and scikit-learn) for analysis and machine learning, and Tableau for interactive dashboards.

Primary Question	Secondary Question
Which cities have the worst air quality, and what factors drive those differences?	Can we predict PM2.5 concentrations using weather conditions and other pollutant levels?

We build baseline and improved regression models to predict PM2.5.

2. Dataset Description

The dataset contains 10,000 records of air quality measurements from 20 major cities across 19 countries, recorded throughout 2023. Each record includes six pollutant concentrations and three meteorological variables.

10,000	20	12	2023
Records	Cities	Columns	Date Range

Feature Reference

Feature	Type	Unit	Range
PM2.5	Float	µg/m³	5.02 - 149.98
PM10	Float	µg/m³	10.00 - 200.00
NO2	Float	ppb	5.00 - 100.00
SO2	Float	ppb	2.01 - 49.99
CO	Float	ppm	0.10 - 9.99
O3	Float	ppb	10.01 - 199.99
Temperature	Float	°C	-10.0 - 40.0
Humidity	Float	%	10.01 - 99.99

Wind Speed	Float	km/h	0.50 - 20.00

Table 1: Numeric features in the dataset

3. ETL and Data Cleaning (Excel)

The data cleaning step was performed using Microsoft Excel. The Excel workbook contains five sheets:

- Raw Data (sample)
- Cleaning Log
- Cleaned Data (sample)
- Data Dictionary
- Summary Statistics

Key Cleaning Findings

- No missing values found across any column (0 out of 120,000 cells)
- No exact duplicate rows detected
- All pollutant values were non-negative and within physically plausible ranges
- Date strings converted to proper date format
- Temperature range (−10 to 40°C) includes sub-zero values for tropical cities (e.g., Bangkok), suggesting the dataset is synthetically generated — this was flagged in the cleaning log

Derived Columns Added

- Month (integer 1–12, extracted from Date) for seasonal analysis
- Air Quality Index (AQI) Category (categorical, based on WHO PM2.5 breakpoints: Good · Moderate · Unhealthy for Sensitive Groups · Unhealthy · Very Unhealthy)

4. Data Storage (Python + SQLite)

The cleaned dataset was loaded into a SQLite database (air_quality.db) with two tables:

- air_quality: 10,000 rows, full cleaned data
- city_summary: 20 rows, pre-aggregated city averages

SQLite was chosen because it is file-based (no server setup required) and integrates directly with Python's sqlite3 module.

Demonstrated Queries

- Ranking cities by average PM2.5
- Computing monthly trends
- Identifying cities exceeding the WHO annual PM2.5 guideline of 15 μg/m³

Key Finding

All 20 cities exceed the WHO annual PM2.5 guideline of 15 µg/m³. Top three: Dubai (80.01), Sydney (78.93), and Mumbai (78.90).

5. Exploratory Data Analysis (Python)

EDA was performed using pandas, matplotlib, and seaborn. The analysis covers distributions, correlations, city comparisons, and seasonal patterns.

5.1 Distributions

All numeric features show approximately uniform distributions, which is unusual for real-world environmental data (which typically follows normal or log-normal distributions). This further confirms the dataset is synthetically generated with random values.

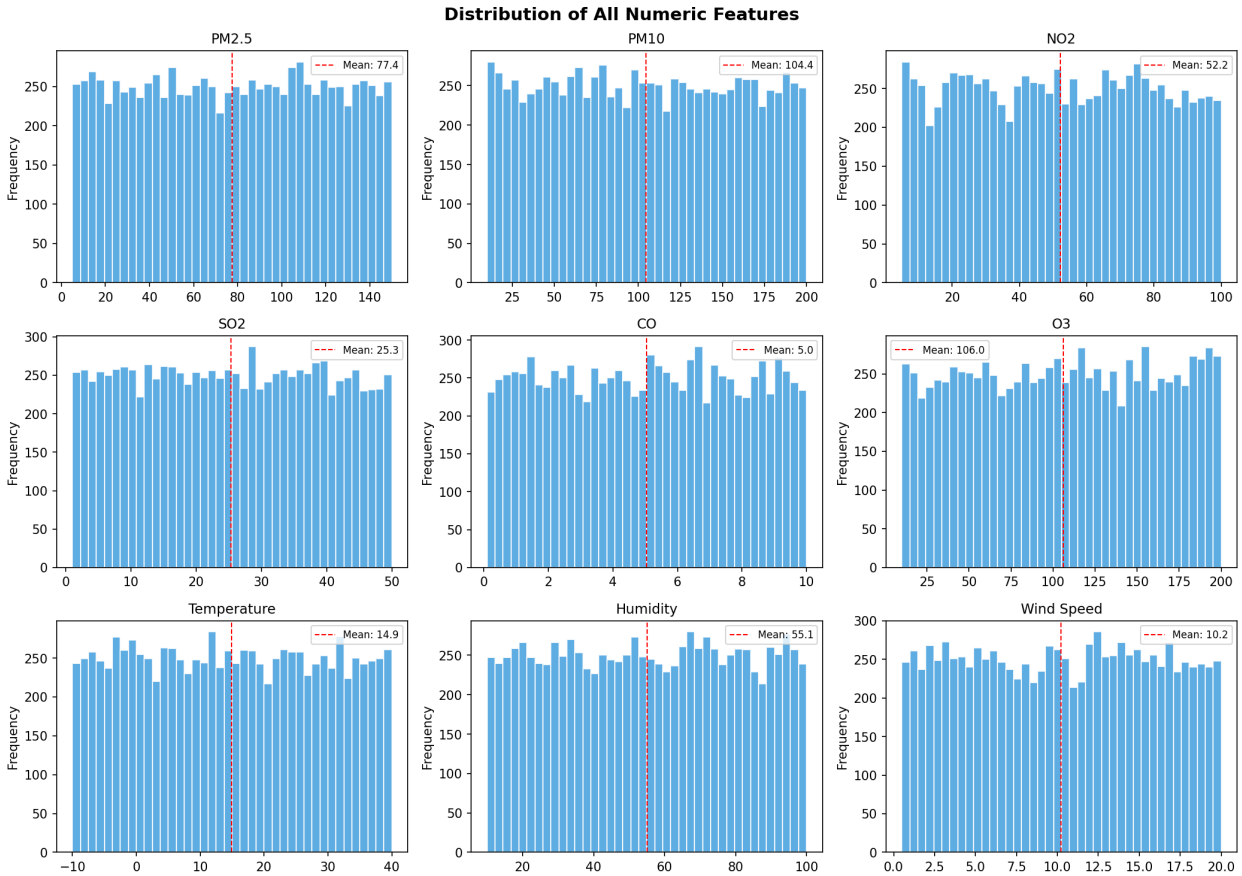

Figure 1: Distribution of all numeric features

5.2 Correlation Analysis

The correlation matrix reveals that all pairwise correlations are extremely weak (|r| < 0.02). PM2.5 shows near-zero correlation with all other features.

Critical Insight

Features are essentially independent random variables. In a real-world dataset, we would expect significant correlations between pollutants (e.g., PM2.5 and PM10) and between weather and pollution (e.g., wind speed inversely correlated with particulate matter). The absence of these correlations will directly limit predictive model performance.

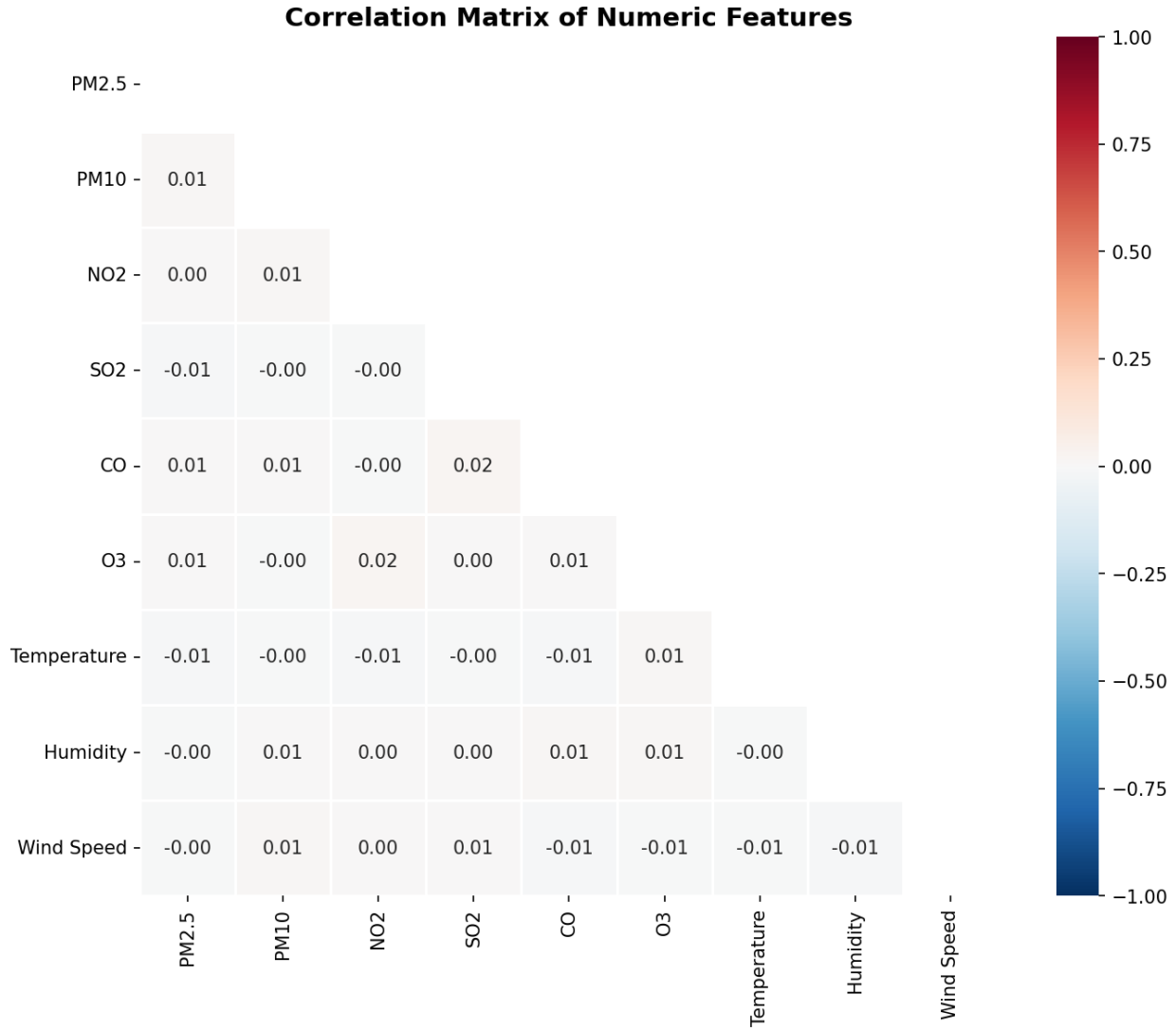

Figure 2: Correlation heatmap of numeric features

5.3 City Comparisons

Average PM2.5 levels range from 74.69 (Paris) to 80.01 (Dubai), a narrow spread of only ~5 μg/m³. All 20 cities far exceed the WHO annual guideline of 15 μg/m³. The small variance between cities is another artifact of the synthetic data generation process.

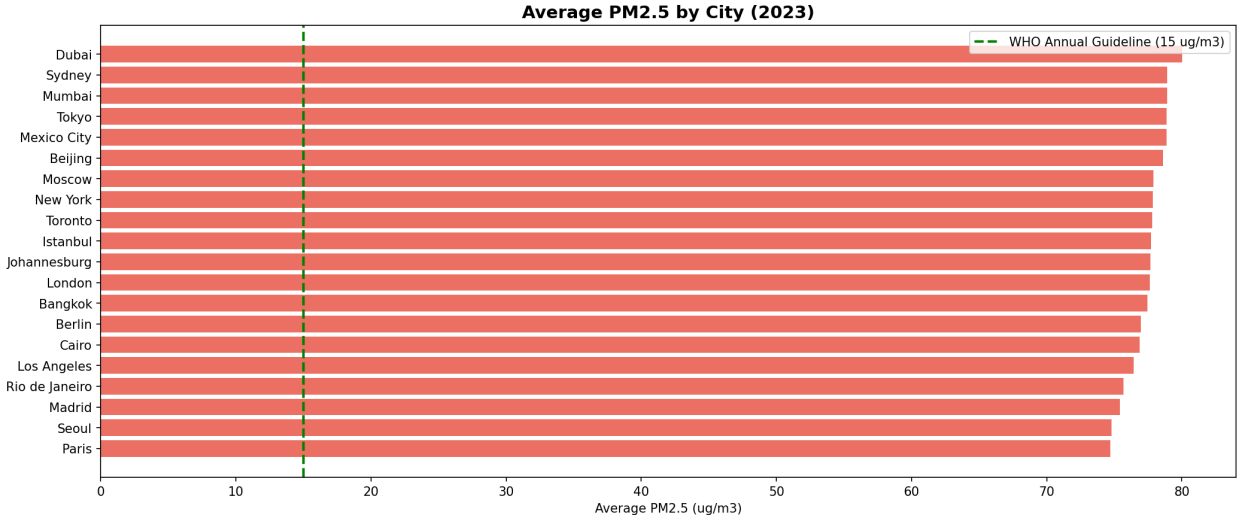

Figure 3: Average PM2.5 by city with WHO guideline reference

5.4 Seasonal Patterns

Monthly averages for PM2.5 and PM10 show no clear seasonal pattern, remaining relatively flat across all months. Temperature and humidity also show minimal monthly variation, which contradicts real-world expectations of seasonal weather patterns.

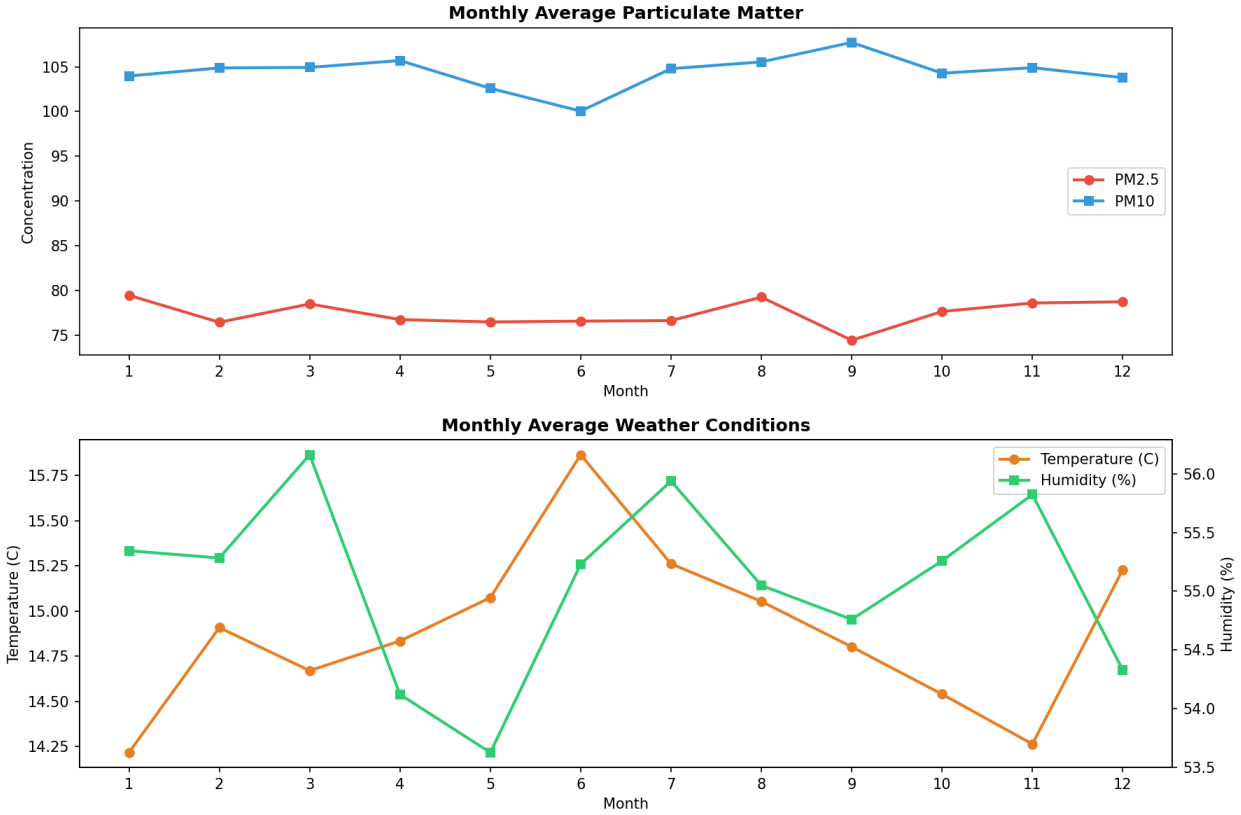

Figure 4: Monthly trends in particulate matter and weather conditions

EDA Summary

Near-zero correlations between features confirm this is a synthetic dataset with independently generated columns. This insight directly informs the prediction task: we expect low model performance regardless of algorithm choice, and this is acknowledged as a core limitation.

6. Prediction Model (Python + scikit-learn)

The applied task is to predict PM2.5 concentrations using the remaining features (other pollutants, weather conditions, city, and month). We used an 80/20 train/test split with a fixed random seed for reproducibility.

Model Results

Metric	Linear Regression (Baseline)	Random Forest (Improved)
MAE	36.08	36.15
RMSE	41.91	42.11
R-squared	−0.0023	−0.0115

Table 2: Model performance comparison

Interpretation

Both models produce negative R-squared values (worse than predicting the mean), confirming that the features have no predictive power for PM2.5 in this synthetic dataset. The MAE of ~36 µg/m³ is close to the standard deviation of PM2.5 (~42 µg/m³), which is expected from near-random predictions. This outcome was anticipated from the EDA correlation analysis.

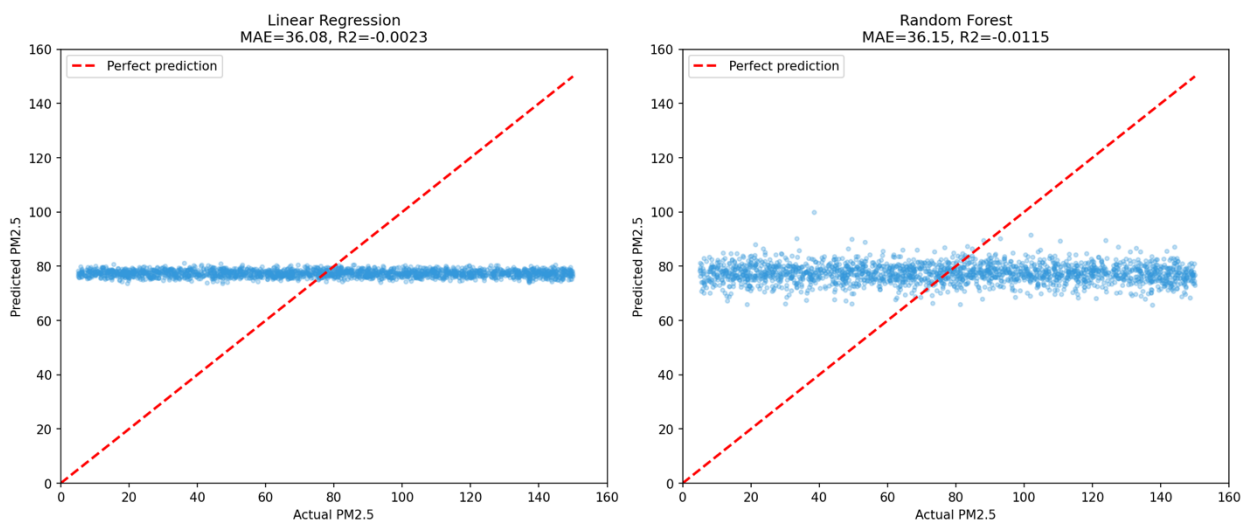

Figure 5: Actual vs. Predicted PM2.5 for both models

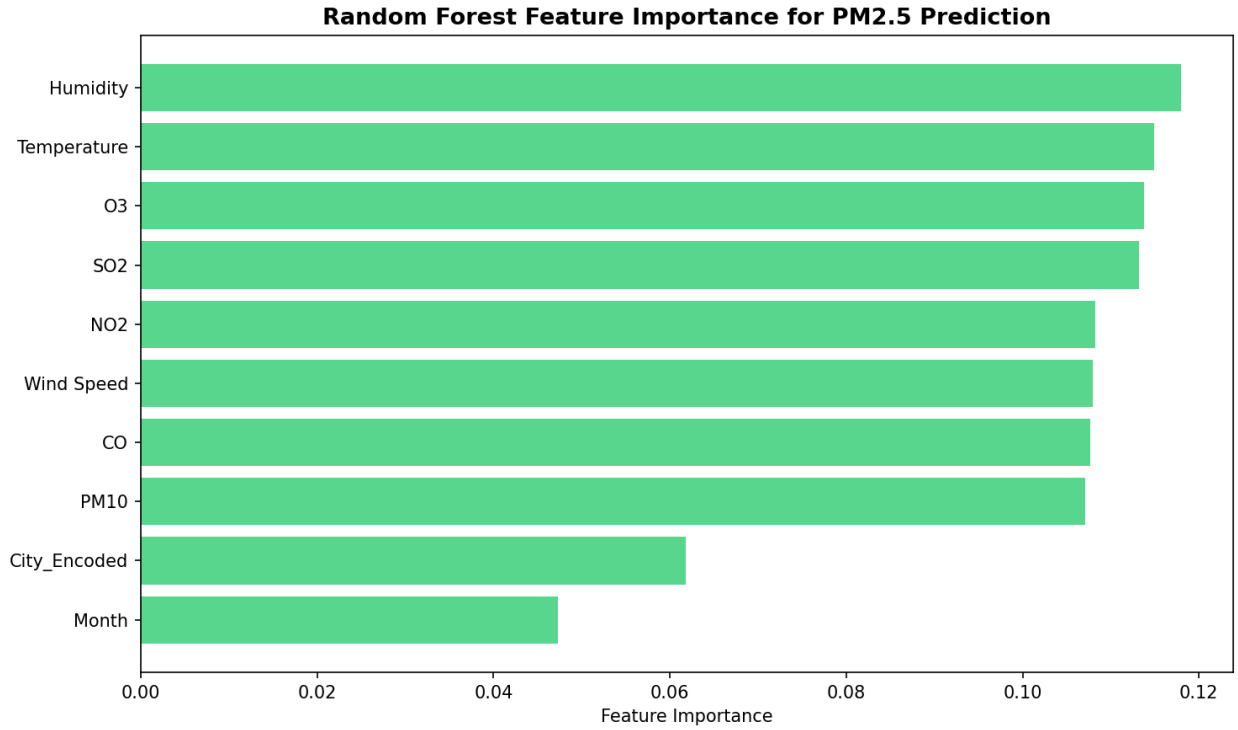

Figure 6: Random Forest feature importance

7. Visualization and Dashboard (Tableau)

An interactive Tableau dashboard was designed with four views:

- City PM2.5 comparison bar chart with WHO guideline reference line
- Monthly pollution trend lines
- Pollutant heatmap by city
- AQI category distribution chart

The dashboard includes a city filter for interactive exploration.

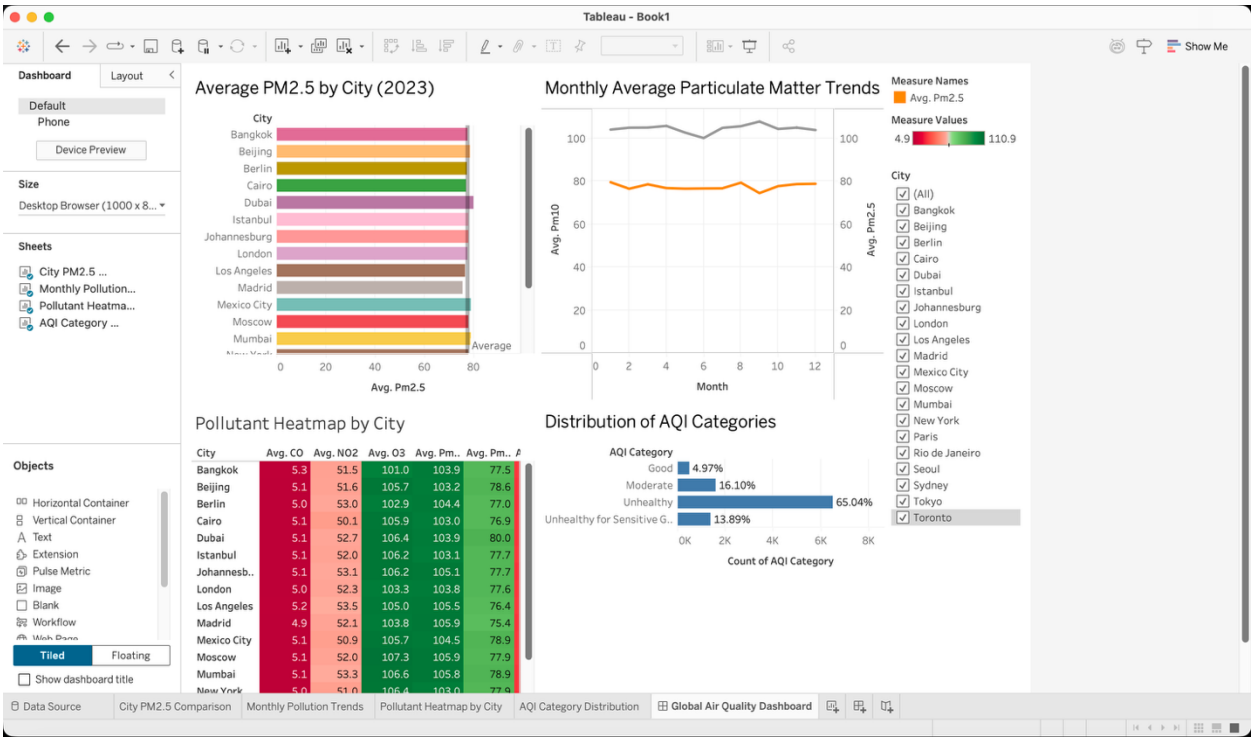

Figure 7: Tableau Dashboard

8. Results and Discussion

All 20 cities in the dataset exceed the WHO annual PM2.5 guideline of 15 µg/m³, with average concentrations ranging from 74.69 (Paris) to 80.01 (Dubai). However, the narrow spread between cities (~5 µg/m³) and the lack of correlation between features strongly suggest the dataset is synthetically generated with independent random distributions.

The prediction models achieved negative R-squared values, meaning they perform worse than simply predicting the mean PM2.5. This is consistent with the EDA finding that features are uncorrelated. While the models themselves are correctly implemented (as demonstrated by the methodology), the dataset does not contain the signal needed for meaningful predictions.

Project Outcome

Despite the synthetic nature of the data, this project successfully demonstrates the complete analytics pipeline methodology: ETL with documentation, structured storage, rigorous EDA, baseline-to-improved modeling with proper evaluation, and multi-tool reporting.

9. Limitations and Next Steps

Limitations

- The dataset is synthetic, with uniformly distributed features and no inter-variable correlations, limiting real-world applicability
- Temperature values are unrealistic for some cities (e.g., sub-zero for Bangkok)
- Only 20 cities are represented, limiting geographic generalizability
- AQI categories are based solely on PM2.5; real AQI calculations consider multiple pollutants

Next Steps

- Replace the dataset with real air quality data from OpenAQ or the EPA Air Quality System
- Implement time-series forecasting to predict future pollution levels
- Add geographic analysis to visualize pollution on a world map
- Expand cloud integration with automated ETL pipelines using Google Cloud Dataflow

10. References

- Waqar, Ali. (2023). Global Air Quality [Dataset]. Kaggle. https://www.kaggle.com/datasets/waqi786/global-air-quality-dataset
- CC0 1.0 Universal (Public Domain Dedication). Open Data Commons.
- https://hdr.undp.org/data-center/human-development-index#/indicies/HDI
- https://data.worldbank.org/indicator/NY.GDP.PCAP.CD